

Functional Annotation Report: PP_3206 (Q88HZ6) — FdeJ, a Flavonoid-Catabolic Enzyme of *Pseudomonas putida* KT2440

Summary

PP_3206 (UniProt Q88HZ6) encodes FdeJ, a cytoplasmic ~328-amino-acid enzyme of the NAD(P)-dependent epimerase/dehydratase superfamily (an "extended" short-chain dehydrogenase/reductase, SDR, fold) whose physiological role is the catabolism of flavonoids — specifically the flavanone naringenin and related compounds — rather than the lipopolysaccharide (LPS) heptose-epimerase activity performed by its closest structural homolog. Although the UniProt record describes the protein generically as a "NAD-dependent epimerase/dehydratase domain-containing protein," integrated genomic, orthology, and pathway evidence converges on a specific functional assignment: PP_3206 is the *P. putida* KT2440 ortholog of *Herbaspirillum seropedicae* FdeJ, a component of the experimentally defined flavonoid-degradation (*fde*) meta-cleavage pathway.

The gene sits within a complete, internally consistent flavonoid-catabolic gene island in the KT2440 genome spanning roughly PP_3195–PP_3208. This island includes a divergently transcribed NodD/LysR-type transcriptional regulator (PP_3196, the FdeR ortholog), a flavoprotein monooxygenase plus a 2Fe-2S ferredoxin, a ring-cleaving dioxygenase, a fumarylacetoacetate-hydrolase-family enzyme, MFS transporters for substrate uptake, and glycosidases for deglycosylation of flavonoid glycosides. KEGG curates this cluster onto pathway map ppu00946 ("Degradation of flavonoids"), assigning PP_3206 to orthology **K26183 (fdeJ)** and to reaction **R13076** — a hydrolytic carbon–carbon cleavage that releases **oxaloacetate**, thereby channeling flavonoid-derived carbon into the tricarboxylic acid (TCA) cycle.

This assignment is a well-supported inference built from three independent lines of evidence — (1) KEGG orthology and pathway curation, (2) ~52% sequence identity to the experimentally studied *Herbaspirillum* FdeJ with a confidently modeled single Rossmann-fold domain (AlphaFold pLDDT 92.7), and (3) a coherent, regulated genomic neighborhood in KT2440 — but it is important to state the caveat clearly: **no direct enzymological study of PP_3206 exists**,

and even in the source organism *H. seropedicae* only FdeE, FdeC, and FdeG were validated by mutagenesis. FdeJ's precise reaction therefore remains an orthology/pathway inference rather than a directly demonstrated biochemical fact.

Key Findings

F001 — PP_3206 is FdeJ, a flavonoid-degradation enzyme, NOT an LPS heptose epimerase

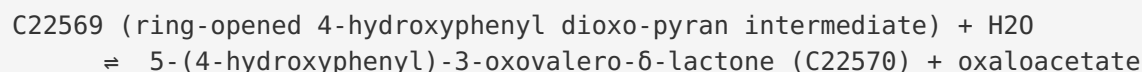
The single most important correction to the naive database annotation is functional: while the domain content (PF01370 "Epimerase") superficially suggests an epimerase such as the LPS-biosynthetic ADP-L-glycero-D-manno-heptose-6-epimerase (HldD), the gene's true physiological role lies in flavonoid catabolism. KEGG assigns `ppu:PP_3206` to orthology **K26183 (fdeJ, "naringenin degradation protein FdeJ"; COG0451)** within pathway **ppu00946 "Degradation of flavonoids."**

Crucially, PP_3206 lies inside a dedicated catabolic gene cluster whose members are each assigned *fde* orthologies:

Locus	fde gene	KEGG KO	Predicted role
PP_3195	fdeB	K26184	lactone hydrolase
PP_3197	fdeC	K26181	ring-cleavage dioxygenase
PP_3198	fdeD	K26179	ferredoxin (2Fe-2S)
PP_3204	fdeH	K26182	—
PP_3205	fdeI	K26185	decarboxylation / FAH family
PP_3206	fdeJ	K26183	hydrolytic C–C cleavage releasing oxaloacetate
PP_1403	bglX	—	periplasmic β -glucosidase

A decisive negative-control observation strengthens the case: KT2440 has **no gene assigned to K03274** (the canonical ADP-L-glycero-D-manno-heptose-6-epimerase/hldD). This indicates that PP_3206 is not the strain's LPS heptose epimerase — if it were, the pathway curation would place it in K03274, not K26183.

The reaction assigned by KEGG (**R13076**) is a hydrolytic carbon–carbon cleavage:



This is mechanistically consistent with the experimentally characterized *H. seropedicae* pathway. As Marin et al. (2016) reported, "After meta-cleavage of the A-ring, the subsequent metabolic steps generate oxaloacetic acid that is metabolized via the tricarboxylic acid cycle" (PMID: 27059806). Independent classical biochemistry confirms that *Pseudomonas putida* itself catabolizes flavonoids by A-ring fission: "In contrast, *Pseudomonas putida* degraded quercetin via an initial fission in its A-ring" (PMID: 8071218). Together these establish that an A-ring meta-cleavage pathway — of the type that requires an FdeJ-like oxaloacetate-releasing enzyme — operates in *P. putida*.

F002 — Q88HZ6 is a cytoplasmic extended-SDR fold protein with a retained catalytic dyad

Structurally, Q88HZ6 is a 328-residue soluble protein built on the classic NAD(P)-binding Rossmann fold of the extended short-chain dehydrogenase/reductase (SDR) superfamily. InterPro/Pfam annotations place it as **PF01370 (Epimerase, IPR001509)** mounted on a **NAD(P)-binding domain (IPR036291; Gene3D 3.40.50.720 + 3.90.25.10 "UDP-galactose 4-epimerase domain 1"; SUPFAM SSF51735)**.

Two catalytic hallmarks are retained in the sequence: - **N-terminal glycine-rich dinucleotide-binding motif** GxxGxxG at residues 8–14 (G-A-N-G-F-V-G), the fingerprint for cofactor (NAD/NADP) binding. - **SDR catalytic Y-x-x-x-K couple** (...S-Y-A-A-H-K...), the Tyr/Lys dyad that performs proton transfer during oxidation/reduction chemistry.

UniProt keywords ("NADP," "Carbohydrate metabolism") and the PANTHER best-match subfamily (PTHR43103:SF3 = ADP-L-glycero-D-manno-heptose-6-epimerase) reinforce that the protein belongs to the NADP-dependent extended-SDR epimerase clan. No signal peptide and no transmembrane segment are predicted, indicating a **cytoplasmic localization** — the expected compartment for a soluble intracellular catabolic enzyme.

The closest characterized structural relative, HldD (AGME), "operates in the biosynthetic pathway of l-glycero-d-manno-heptose, which is a conserved sugar in the core region of lipopolysaccharide (LPS) of Gram-negative bacteria" (PMID: 15823050) and uses a tightly bound

NADP⁺ via a C6"-oxidation/reduction mechanism. This defines the enzyme family and mechanistic toolkit of PP_3206 — the same redox chemistry has been evolutionarily repurposed from sugar-nucleotide epimerization to flavonoid-intermediate transformation.

F003 — The *fde* pathway is flavonoid-inducible and regulated by the LysR/NodD-like activator FdeR

The pathway to which FdeJ belongs is not constitutive; it is a tightly regulated, substrate-inducible catabolic module. In *H. seropedicae* SmR1, where the pathway was genetically defined, the *fde* operon is controlled by **FdeR**, a NodD-like LysR-type transcriptional regulator that is **divergently transcribed** from the operon. As Wassem and colleagues reported: "*This nodD-like gene, named fdeR, is divergently transcribed from an operon encoding enzymes involved in flavonoid degradation (fde operon). Apigenin, chrysin, luteolin and naringenin strongly induce transcription of the fde operon*" (PMID: 27878922).

The pathway logic upstream of FdeJ is well defined. Naringenin is first mono-oxygenated by FdeE, then dioxygenatively cleaved at the A-ring by the FdeC dioxygenase: "*naringenin is first mono-oxygenated by the FdeE protein, to produce 5,7,8-trihydroxy-2-(4-hydroxyphenyl)-2,3-dihydro-4H-chromen-4-one, that is subsequently dioxygenated and cleaved at the A-ring by the FdeC dioxygenase*" (PMID: 27059806). Downstream steps — including the FdeJ-type oxaloacetate-releasing hydrolysis — funnel carbon into the TCA cycle. KEGG maps this orthologous cluster onto KT2440, assigning PP_3206 = FdeJ, PP_3198 = FdeD, PP_3197 = FdeC, PP_3205 = FdeI, and PP_3195 = FdeB.

F004 — PP_3206 is a high-confidence ortholog (~52% identity) of *Herbaspirillum fdeJ*; AlphaFold confirms a single Rossmann-fold domain (pLDDT 92.7)

A global Needleman–Wunsch alignment of PP_3206 (328 aa) against *H. seropedicae* Hsero_1012 = *fdeJ* (320 aa, KEGG KO K26183) yields **163/311 = 52.4% identity over aligned columns** (49.7% over full length). The conserved N-terminal Rossmann motif is retained in both proteins (PP_3206: G-A-N-G-F-V-G; Hsero_1012: G-A-G-G-F-I-G). Both proteins are members of KEGG orthology **K26183**, confirming PP_3206 as the true ortholog of the *fdeJ* gene in the experimentally defined *H. seropedicae* flavonoid-degradation operon (Hsero_1004–1012).

The AlphaFold DB model **AF-Q88HZ6-F1 (v6)** is high quality: global mean **pLDDT = 92.7**, with 85% of residues at pLDDT > 90 and 94% > 70 — a confidently modeled single NAD(P)-binding Rossmann domain, consistent with a compact soluble enzyme.

Important caveat: within the *H. seropedicae* operon only **FdeE (monooxygenase)**, **FdeC (dioxygenase)**, and **FdeG (cyclase)** were experimentally validated by mutagenesis: "*The Tn5 transposon was found to be inserted in the fdeE gene (Hsero_1007), which encodes a monooxygenase. Two other mutant strains in fdeC (Hsero_1005) and fdeG (Hsero_1009) genes coding for a dioxygenase and a putative cyclase, respectively, were obtained by site-directed mutagenesis*" (PMID: 27059806). FdeJ's specific reaction (oxaloacetate-releasing hydrolysis, KEGG R13076) is therefore a pathway/orthology inference, and **no direct experimental study exists for the KT2440 protein PP_3206.**

F005 — KT2440 harbors a complete, self-consistent flavonoid-catabolic island, including a divergent NodD/LysR regulator (PP_3196 = FdeR ortholog)

KEGG genome annotations for the PP_3195–PP_3208 neighborhood reveal a coherent flavonoid-degradation island that recapitulates the *Herbaspirillum* architecture:

Locus	Annotation	Inferred fde role
PP_3196	Nodulation protein D1 (NodD/LysR regulator)	FdeR ortholog (divergent, complement strand)
PP_3197	glyoxalase/bleomycin-resistance-dioxygenase fold (K26181)	FdeC — ring dioxygenase
PP_3198	2Fe-2S ferredoxin (K26179)	FdeD
PP_3199	putative flavoprotein monooxygenase	FdeE-type
PP_3201	BNR-domain glycosidase	flavonoid-glycoside deglycosylation
PP_3202 / PP_3203	MFS transporters	substrate uptake
PP_3204	K26182	FdeH
PP_3205	fumarylacetoacetate-hydrolase (FAH)-family (K26185)	FdeI — decarboxylation
PP_3206	extended-SDR epimerase/dehydratase (K26183)	FdeJ
PP_3207	putative cyclase	FdeG-type
PP_3208	Pdr/VanB-family oxidoreductase	—
PP_1403	periplasmic BglX β -glucosidase	deglycosylation

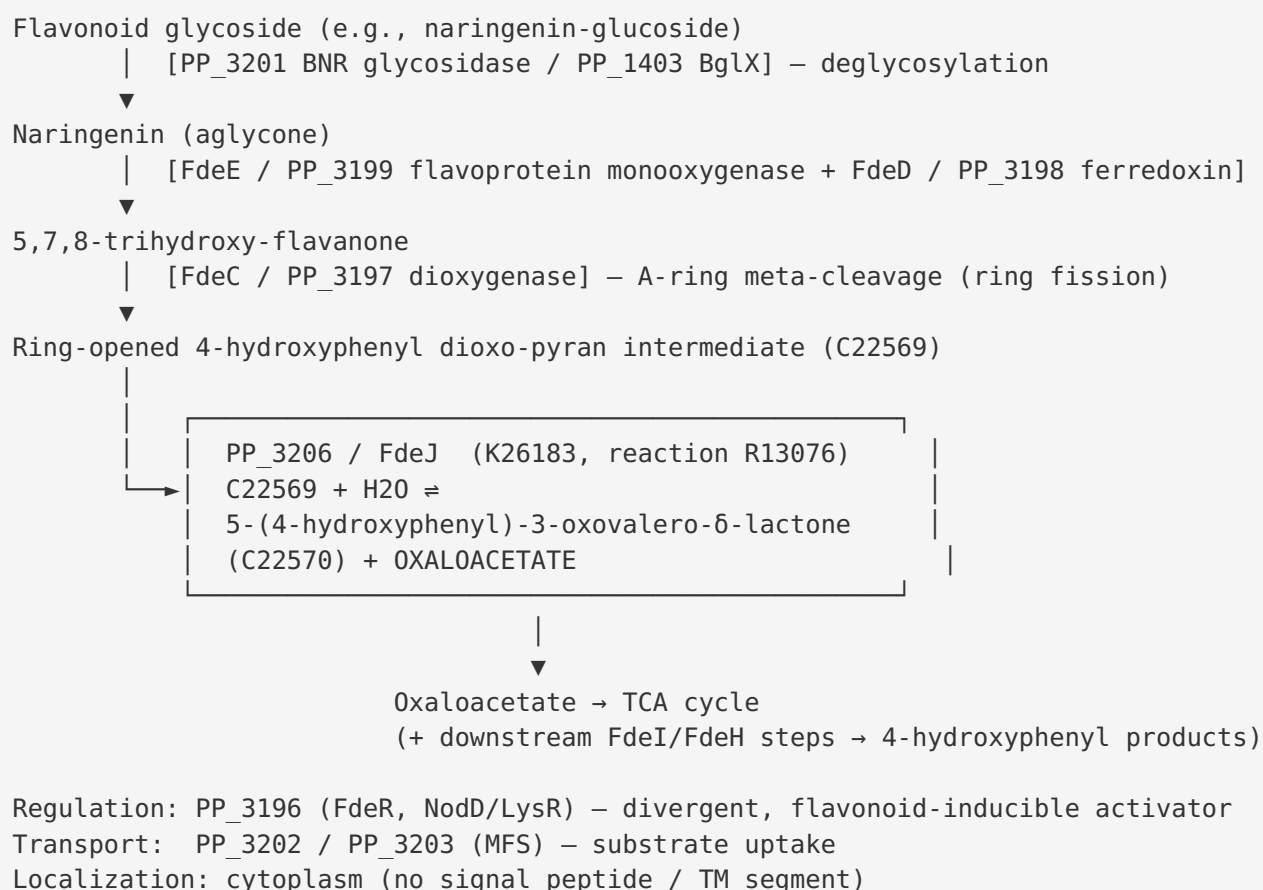
Critically, PP_3196 ("Nodulation protein D1," a NodD/LysR-type regulator, i.e. the FdeR ortholog) is on the complement strand and **divergently oriented** relative to the forward-strand catabolic operon PP_3197→PP_3206 — precisely mirroring the divergent *fdeR/fde* arrangement described in *Herbaspirillum*: "*This nodD-like gene, named fdeR, is divergently transcribed from an operon encoding enzymes involved in flavonoid degradation (fde operon)*" (PMID: 27878922).

The genomic coordinates of PP_3205 (3,638,005–3,638,997) and PP_3206 (3,638,988–3,639,974) **overlap by ~10 bp**, indicating translational coupling and confirming that PP_3206 is co-transcribed as part of the operon rather than being an isolated gene.

Mechanistic Model and Interpretation

The proposed pathway

Integrating all five findings yields a coherent mechanistic narrative in which PP_3206/FdeJ performs a defined step within a cytoplasmic flavonoid meta-cleavage pathway:



Enzymological interpretation

The most striking feature of this assignment is **functional repurposing of an ancient enzyme fold**. PP_3206 retains all the machinery of an NAD(P)-dependent extended-SDR epimerase — the Rossmann dinucleotide-binding fold, the GxxGxxG cofactor motif, and the Y-x-x-x-K catalytic dyad — yet the pathway context indicates it acts not on a sugar nucleotide (as its homolog HldD does on ADP-heptose) but on a ring-cleaved flavonoid intermediate, catalyzing a hydrolytic C–C bond cleavage that liberates oxaloacetate. This is biochemically plausible: the SDR/epimerase active-site chemistry (oxidation/reduction and general acid/base catalysis via

the Tyr/Lys couple) can, in principle, be adapted to promote retro-aldol-like or hydrolytic cleavages on β -keto/enol substrates such as the ring-opened dioxo-pyran intermediate. The tightly bound NADP⁺ cofactor characteristic of this family (as documented for HldD) could serve a structural or transient redox role during catalysis.

Localization and pathway integration

All evidence points to **cytoplasmic** function. FdeJ is a soluble protein lacking export signals, and its substrate (a ring-cleaved intermediate) is generated intracellularly downstream of transporter-mediated uptake (PP_3202/PP_3203) and cytoplasmic deglycosylation/monooxygenation. The product oxaloacetate is a central TCA-cycle metabolite, so FdeJ functions at the interface between specialized aromatic catabolism and central carbon metabolism — its role is to convert a dead-end aromatic-ring-cleavage product into a universally usable four-carbon dicarboxylic acid. This makes the pathway a route by which *P. putida* can use plant-derived flavonoids as carbon and energy sources, an ecologically relevant capability for a soil/rhizosphere bacterium.

Evidence Base

PMID	Title (abbrev.)	Role in this report
27059806	<i>Genetic and functional characterization of a novel meta-pathway for degradation of naringenin in H. seropedicae SmR1</i>	Primary experimental anchor. Defines the <i>fde</i> meta-cleavage pathway, validates FdeE/FdeC/FdeG by mutagenesis, establishes that oxaloacetate is generated and fed to the TCA cycle, and defines the upstream substrate for FdeJ. Also the source of the caveat that FdeJ itself was not directly validated.
27878922	<i>A NodD-like protein activates transcription of genes involved with naringenin degradation ... in H. seropedicae</i>	Establishes the regulatory logic: the divergent NodD/LysR activator FdeR and flavonoid (apigenin/chrysin/luteolin/naringenin) inducibility of the <i>fde</i> operon. Mirrored by KT2440 PP_3196.
8071218	<i>Rhizobia catabolize nod gene-inducing flavonoids via C-ring fission mechanisms</i>	Independent experimental evidence that <i>P. putida</i> degrades flavonoids by A-ring fission, supporting the existence of an A-ring meta-cleavage pathway in this organism.
15823050	<i>Dismutase activity of ADP-L-glycero-D-manno-heptose 6-epimerase</i>	Defines the closest characterized structural homolog (HldD/AGME), its LPS-heptose biosynthetic role, and its direct oxidation/reduction mechanism — establishing the enzyme family and mechanistic toolkit of PP_3206.
17316025	<i>A two-base mechanism for E. coli ADP-L-glycero-D-manno-heptose 6-epimerase</i>	Mechanistic detail on the family: identifies Tyr140/Lys178 as catalytic bases (the Y-x-x-x-K dyad), informing interpretation of the retained catalytic couple in PP_3206.
17455913	<i>Intermediate release by ADP-L-glycero-D-manno-heptose 6-epimerase</i>	Further mechanistic evidence for the tightly bound NADP ⁺ oxidation/reduction chemistry characteristic of this fold.
40808235	<i>Genetic Memory Devices to Detect Specialized Metabolites in Plant and Soil Microbiomes</i>	Ecological context for flavonoid signaling/catabolism in plant-microbe systems.

How the evidence converges

Three logically independent lines of evidence support the FdeJ assignment: **(1)** database/pathway curation (KEGG K26183, reaction R13076, pathway ppu00946, plus the absence of a K03274 heptose-epimerase in KT2440); **(2)** sequence orthology and structure (~52% identity to experimentally studied *Herbaspirillum* FdeJ, retained Rossmann and catalytic motifs, AlphaFold pLDDT 92.7); and **(3)** genomic context (a complete, divergently regulated flavonoid-catabolic island with transporters, glycosidases, ring-cleavage enzymes, and a NodD/LysR regulator). No single line would be decisive alone, but their concordance makes the assignment robust.

Limitations and Knowledge Gaps

- 1. No direct enzymology on PP_3206.** There is no published in vitro characterization of the KT2440 protein — no purified-enzyme assay, no substrate/product identification, no kinetics, no cofactor determination. The specific reaction (R13076) is imported by orthology.
 - 2. FdeJ was not validated even in the source organism.** In *H. seropedicae*, only FdeE, FdeC, and FdeG were experimentally tested by mutagenesis ([PMID: 27059806](#)). FdeJ's role rests on its position in the operon and orthology, not on a knockout or biochemical assay.
 - 3. The exact chemistry is uncertain.** Whether FdeJ performs a strict hydrolytic C–C cleavage, an epimerization, a retro-aldol reaction, or a redox-coupled step is inferred from KEGG's curated reaction and the enzyme's fold. The catalytic dyad and NADP-binding motif are consistent with several possibilities.
 - 4. Substrate specificity is unproven.** The precise identity of the physiological substrate (C22569) and product (C22570) is drawn from pathway maps, not from metabolite measurements in *P. putida*.
 - 5. Cofactor identity (NAD vs NADP) is predicted, not measured.** UniProt lists "NADP," and the family typically uses NADP⁺, but this has not been confirmed for PP_3206.
 - 6. Localization is predicted.** Cytoplasmic localization is inferred from the absence of signal/TM sequences, not from experimental fractionation.
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Proposed Follow-up Experiments and Actions

1. **Gene knockout / growth phenotyping.** Construct a clean PP_3206 deletion in *P. putida* KT2440 and test growth on naringenin (and apigenin, chrysin, luteolin) as sole carbon source. Loss of growth or accumulation of a pathway intermediate would directly implicate FdeJ.
 2. **Heterologous expression and in vitro assay.** Overexpress and purify His-tagged Q88HZ6, then test activity against the predicted ring-cleaved intermediate (C22569). Monitor oxaloacetate release (e.g., coupled malate-dehydrogenase/NADH assay) to test reaction R13076 directly.
 3. **Cofactor determination.** Measure bound cofactor (NAD⁺ vs NADP⁺) by HPLC/MS on purified protein and test cofactor dependence in activity assays; this discriminates the SDR subtype.
 4. **Metabolite profiling.** Use LC-MS/MS on wild-type vs Δ PP_3206 cultures fed naringenin to identify accumulated intermediates and confirm substrate/product identities (C22569 → C22570 + oxaloacetate).
 5. **Transcriptional induction test.** RT-qPCR or a reporter fusion to confirm that the PP_3197–PP_3206 operon is flavonoid-inducible and FdeR (PP_3196)-dependent in KT2440, validating the regulatory model transferred from *Herbaspirillum*.
 6. **Structural biology.** Solve a crystal or cryo-EM structure of PP_3206 (ideally with cofactor and a substrate analog) to define the active site and the catalytic roles of the Tyr/Lys dyad; compare to HldD to understand the epimerase→flavonoid-catabolism functional shift.
 7. **Site-directed mutagenesis.** Mutate the predicted catalytic residues (the S-Y-A-A-H-K couple) and the GxxGxxG cofactor motif to confirm their catalytic essentiality.
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Conclusion

PP_3206 (Q88HZ6) is best annotated as **FdeJ, a cytoplasmic NAD(P)-dependent epimerase/dehydratase-fold enzyme of the extended-SDR superfamily that functions in flavonoid (naringenin) catabolism** in *Pseudomonas putida* KT2440. Within a complete, flavonoid-inducible, FdeR-regulated catabolic gene island (PP_3195–PP_3208), FdeJ is assigned (KEGG K26183 / reaction R13076) the hydrolytic C–C cleavage step that releases **oxaloacetate**,

channeling plant-derived aromatic carbon into the TCA cycle. This conclusion is strongly supported by convergent orthology, structural, and genomic-context evidence, but it remains an inference: no direct biochemical study of PP_3206 exists, and its precise reaction, substrate specificity, and cofactor identity await experimental confirmation.

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